

Lec 1-1 (Sept 1, 2025)

- Syllabus
- What is Calculus?
- Rates of Change : Average RC, Instant RC
- Limits

§1. Syllabus

§2. Calculus

§2.1. "Definition"

- Calculus is the study of continuous changes and limits

- Calculus = "calculus of infinitesimal"

- "an amount that gets arbitrarily close to zero"

- "infinite small"

• Calculus is divided into $\left\{ \begin{array}{l} \text{differential calculus} \\ \text{integral calculus} \end{array} \right.$

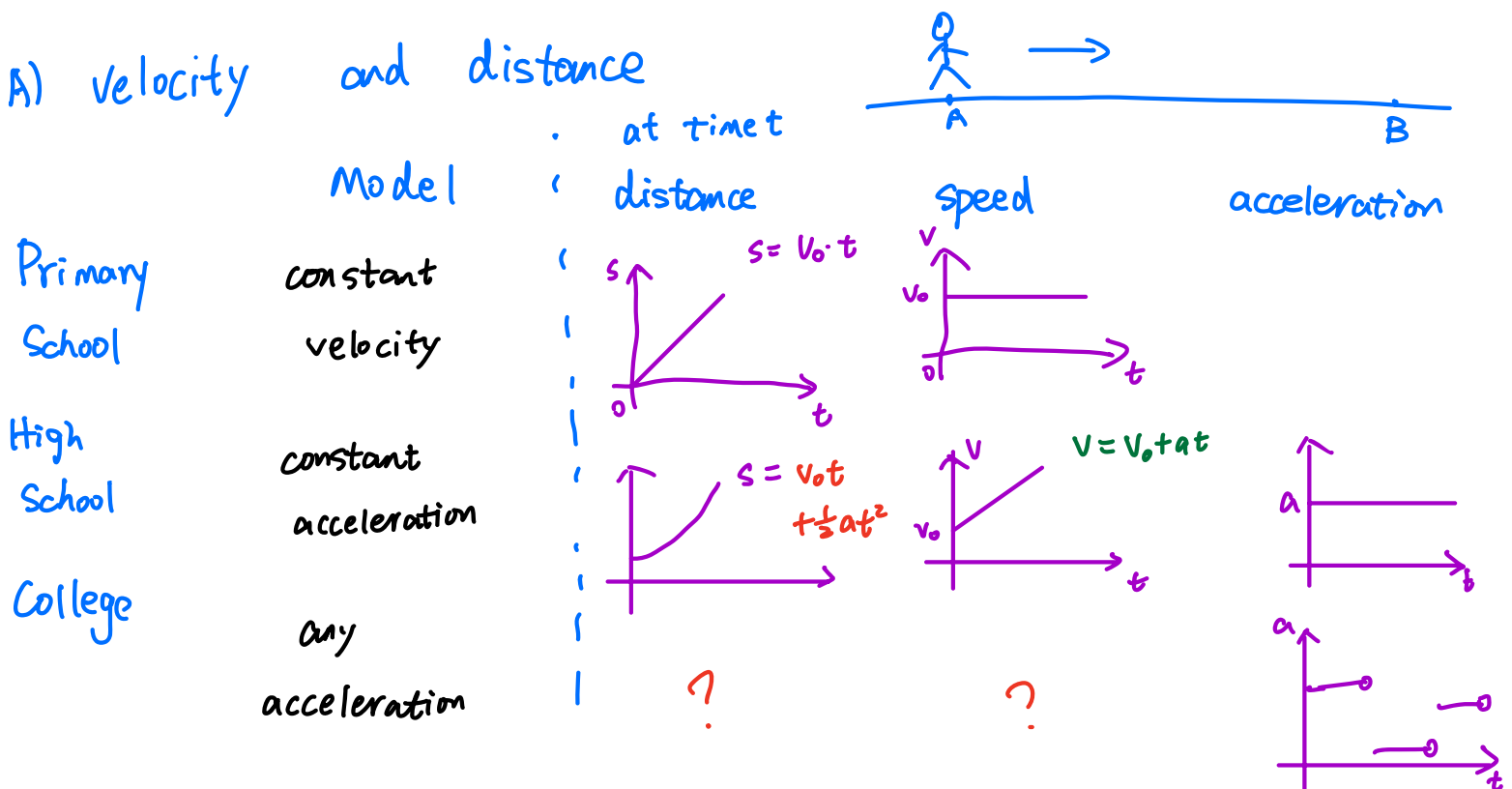
• They are connected by "Fundamental Theorem of Calculus"
(i.e., Newton-Leibniz Theorem)

and rigorously formulated by limits (i.e. the ϵ - δ language).

• In the following, the red part usually means that we will study in the future or it is currently unclear

§ 2.2. Motivating examples

A) velocity and distance



Q1: What is the definition of **acceleration**?

Even more fundamentally, what is the definition of **velocity**?

Q2: How to know the velocity and the distance given a specific a ?

E.g., What is the two "?" above?

B) Areas of shapes.

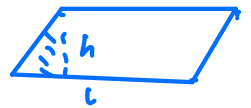
Primary school:



$$S = a \cdot b$$



$$S = h \cdot l \cdot \frac{1}{2}$$

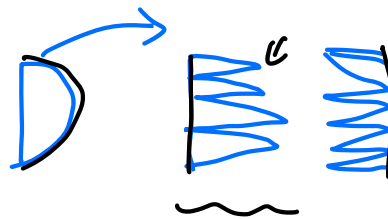


$$S = h \cdot l$$

high school:



$$S = \pi \cdot r^2$$

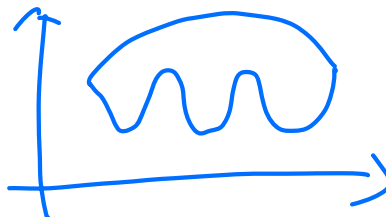


$$S = \pi r \cdot r$$

College:



$$S = ?$$



$$S = ?$$

Q1: Why can we compute the circle area in this "approximation" way? Prove it rigorously.

Q2: How to compute any shape's area given certain curves?

§ 2.3. History [Will not be tested in the exams or assignments]

Ancient times. • Egyptians (1820 BC): areas, lengths

• Archimedes (250 BC): Circle perimeter $= 2\pi r$

E.g. • LIU Hui (AD 280): area of a circle } using division of a circle
• ZU Chongzhi (AD 450): volume of a ball }

1650. • Cavalier: infinitesimal idea

△ Barrow: finite difference $\longleftrightarrow$ infinitesimal

1687. △ Newton

• Leibniz: a system of nice symbolic expressions [dx , $\int$ = sum]
→ combine differential and integral calculus into unified calculus
using fundamental theorem of Calculus [Establishment of Calculus]

- Fermat : Lemma for local minimum [first-order condition]

1715' : • Taylor : series/expansion

- Maclaurin : a special case of Taylor expansion

1750' : • Johann Bernoulli and his family : fast descend line, L'Hopital rule

- Euler : i , π , infinite series

- Lagrange : mean value theorem

1850' : • Cauchy : limit } using ϵ - δ language

- Weierstrass : continuity

- Riemann : integral

1900' : Lebesgue : modern calculus (measure theory)

———— At this point, all rigorousness issues have been done ! [4]

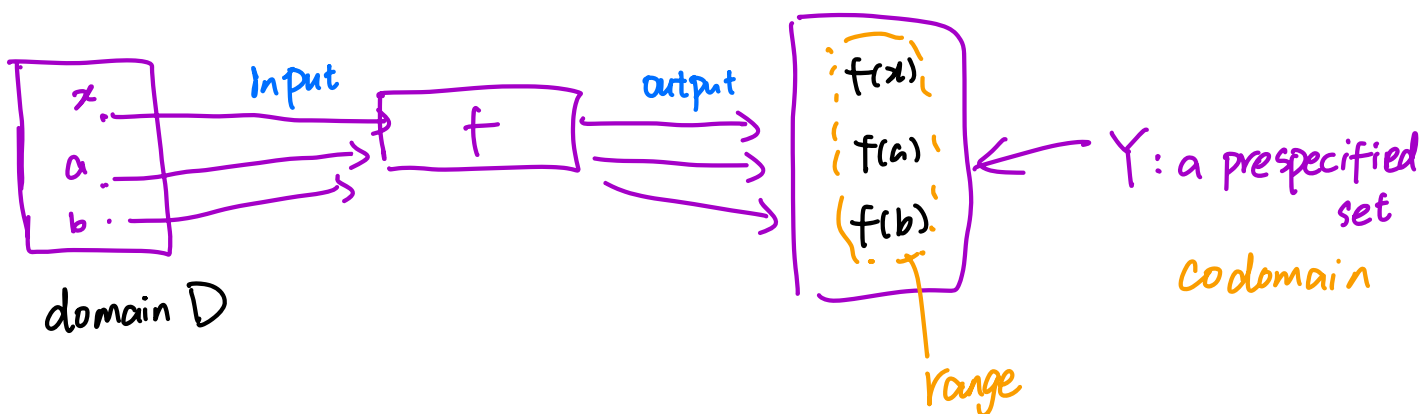
§2.4. Conclusion.

- Wherever there is a change, Calculus can be used to describe the system.
- Friedrich Engels: Calculus is the utmost victory of human spirit.

§3. Rates of change

§3.0. Quick review: Functions

Definition: A function f from a set D to a set Y is a rule that assigns a unique element $f(x) \in Y$ to each element $x \in D$.



§3.2

Notation: $f: D \rightarrow Y$

e.g.1: $f: \mathbb{R} \rightarrow \mathbb{R}$ (the set of all real numbers)
 $x \mapsto x^2$ $\text{Range}(f) = [0, \infty)$

$$f(1) = 1, \quad f(0.1) = 0.01, \quad f(-10) = 100, \dots$$

Definition: The range of a function $f: D \rightarrow Y$ is a set
 $\text{Range}(f) \triangleq \{f(x) : x \in D\}$.

[Here the symbol " $\triangleq$ " or " $:=$ " means "is defined to be".]

Remarks: ① Given $f: D \rightarrow Y$, we have $\text{Range}(f) \subseteq Y$.

② In general, $\text{Range}(f) \neq Y$.

③ Y is a pre-specified set that all the outputs must lie in.

Def. Let f and g be two functions. We have the following operations:

• Plus/sum : $(f+g)(x) \triangleq f(x) + g(x)$

• Difference / Subtraction / minus : $(f-g)(x) \triangleq f(x) - g(x)$

• product / times : $(f \cdot g)(x) \triangleq f(x) \cdot g(x)$

• Division : $(f/g)(x) \triangleq f(x)/g(x)$

• Composition : $(f \circ g)(x) \triangleq \underline{f(g(x))}$
 $x \mapsto \quad \mapsto$

e.g. $f(x) = \sqrt{x}$. $g(x) = \sqrt{1-x}$ $\quad x \geq 0$
 $(f+g)(x) = \sqrt{x} + \sqrt{1-x}$ $\quad 0 \leq x \leq 1$

$(f-g)(x) = \sqrt{x} - \sqrt{1-x}$

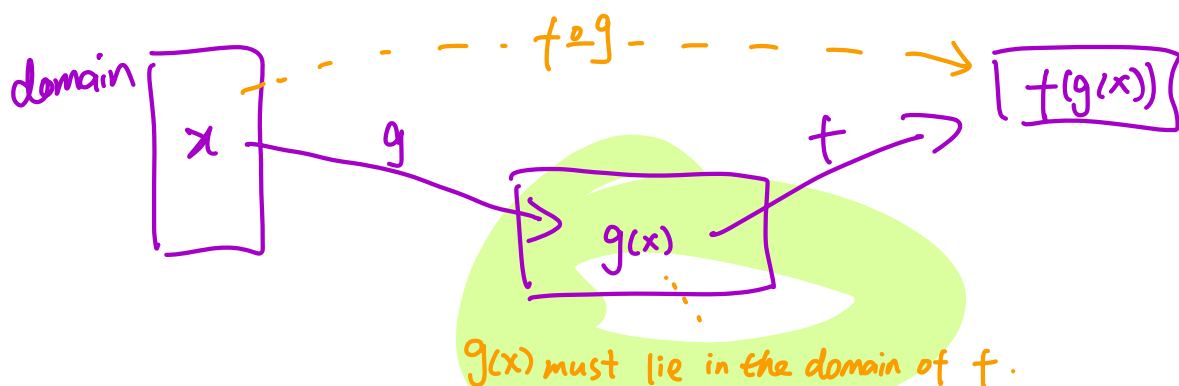
$(f \cdot g)(x) = \sqrt{x} \cdot \sqrt{1-x}$

$(f/g)(x) = \frac{\sqrt{x}}{\sqrt{1-x}}$

$(f \circ g)(x) = \sqrt{\sqrt{1-x}}$
 $x \xrightarrow{g} \sqrt{1-x}$
 $\xrightarrow{f} \sqrt{g(x)}$
 $= \sqrt{\sqrt{1-x}}$

Rmk : ① Make sure that the domains match

E.g., two functions can be composed at x whenever the value of one function at x lies in the domain of the other.



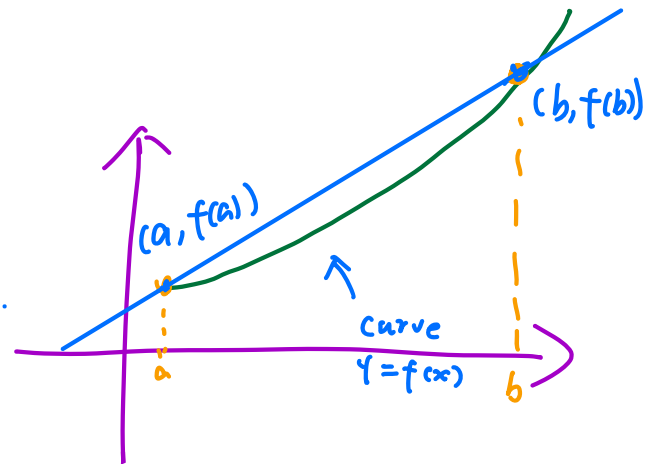
§3.1. Average rates of change (average speed)

Def: Let $y = f(x)$ and suppose $\underline{[a, b]} \subseteq D$ (domain). The average rate of change of y with respect to wrt x over $[a, b]$

is
$$\frac{f(b) - f(a)}{b - a}$$

Def [secant line]

A secant line to a curve is a straight line aligning two distinct points of the curve.



★ Average rate of change of y wrt x over $[a, b]$	$= \frac{f(b) - f(a)}{b - a}$	$=$ the slope of the corresponding secant line.
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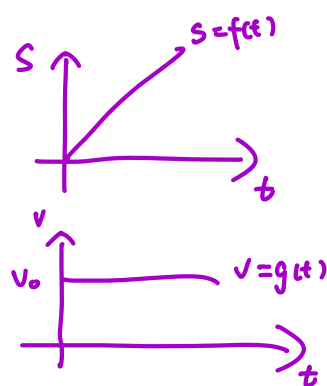
§3.2. Instant Rates of change

"constant velocity model":

$$S = f(t) = v_0 t$$

$$v = g(t) \equiv v_0$$

[" $\equiv$ " means "always equal to".]



Question:

If the distance function is given by $S = f(t) = t^2$,

what is the speed at time $t = 1$? [derive it, not apply it from high school physics directly]

(what is the speed? How to compute?)

concept

computation

① compute the average RC from $t = 1$ to $t = b$.

Idea:

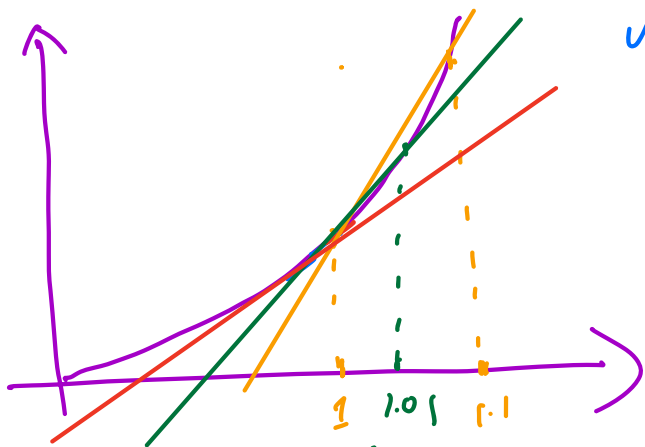
②

	(0.00001)						
b	0.9	0.99	0.999	1	1.001	1.01	1.1
average RC	$\frac{1^2 - 0.9^2}{1 - 0.9}$			?	$\frac{1.001^2 - 1^2}{1.001 - 1}$	$\frac{1.01^2 - 1^2}{1.01 - 1}$	$\frac{1.1^2 - 1^2}{1.1 - 1}$
	= 1.9	= 1.99	1.999		= 2.001	= 2.01	= 2.1

Observe that as b approaches 1, the average speed is getting closer to 2.

We may guess $? = 2$ to be the answer of instant speed.

Further, we show this in a graphic way. We first have secant lines which are respectively given by $[1, 1.1]$ and $[1, 1.01]$. Then as b approaches 1 we obtain the red tangent line.



[An informal definition from Wiki] A tangent line to a curve at a given point is the straight line that "just touches" the curve at that point.

Then, the instant rate of change at time t is defined as the slope of the tangent line. Newton's way:
↙ graphical

To compare, the average rate of change between t and $t+\Delta t$ = the slope of the secant line = $\frac{f(t+\Delta t) - f(t)}{\Delta t}$.

More formally, the instant rate of change of $s=f(t)$ at time t is denoted by

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{f(t+\Delta t) - f(t)}{\Delta t}$$

Leibniz's way:
symbolic.

where Δt is the change in time [independent variable]
and $\Delta s = f(t+\Delta t) - f(t)$ is the change in distance [dependent value].

§3.3. Computation

e.g. For $f(t) = t^2$, we first compute the average speed between $t=1$ and $t=1+\Delta t$:

$$\frac{f(1+\Delta t) - f(1)}{\Delta t} = \frac{(1+\Delta t)^2 - 1^2}{\Delta t} = \frac{1 + 2\Delta t + (\Delta t)^2 - 1}{\Delta t} = 2 + \Delta t.$$

and then compute the instant speed at $t=1$:

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{f(1+\Delta t) - f(1)}{\Delta t} = \lim_{\Delta t \rightarrow 0} (2 + \Delta t) = 2. \quad \square$$

Generally, the average speed between $t+\Delta t$ and t is given by

$$\begin{aligned} \frac{f(t+\Delta t) - f(t)}{\Delta t} &= \frac{(t+\Delta t)^2 - t^2}{\Delta t} = \frac{t^2 + (\Delta t)^2 + 2t \cdot \Delta t - t^2}{\Delta t} \\ &= \frac{(\Delta t)^2 + 2t \cdot \Delta t}{\Delta t} = \Delta t + 2t, \end{aligned}$$

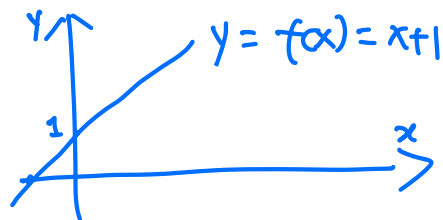
and the instant speed at time t is

$$\lim_{\Delta t \rightarrow 0} \frac{f(t+\Delta t) - f(t)}{\Delta t} = \lim_{\Delta t \rightarrow 0} (\Delta t + 2t) = 2t.$$

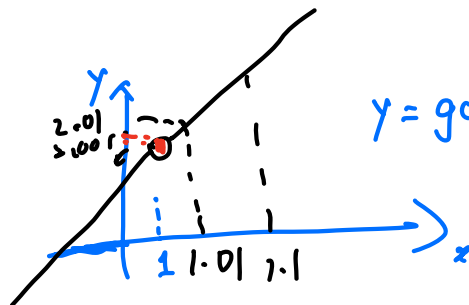
✓
Exercise. I mentioned the constant acceleration model where the distance is given by $s = f(t) = v_0 t + \frac{1}{2} a t^2$. Derive the speed formula at time t by yourself.

§ 4. Limits

§ 4.1. An example.



domain: $(-\infty, \infty)$



$$y = g(x) = \frac{x^2 - 1}{x - 1} = \begin{cases} x + 1, & x \neq 1 \\ \text{DND}, & x = 1 \end{cases}$$

domain: $(-\infty, 1) \cup (1, \infty)$

Difference: The graph of g is identical with the line $f(x) = x + 1$ except at $x = 1$, where g is not define.

We list the same table:

x	0.9	0.99	0.999	1	1.001	1.01	1.1
$g(x)$	1.9	1.99	1.999	?	2.001	2.01	2.1

- Denote $\lim_{x \rightarrow 1} g(x) = 2$, which reflects the "trend" of g at $x = 1$

although g is not define at $x = 1$.

We emphasize first time
that the limit is the trend.

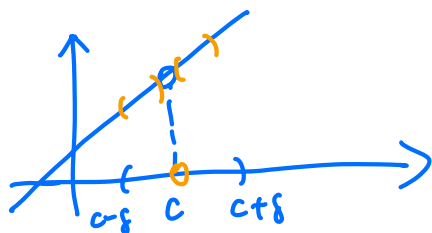
§4.2. Important note : neighborhood without the center

When we discuss $\lim_{x \rightarrow c} f(x)$, we assume that $f: D \rightarrow \mathbb{R}$ is defined in the neighborhood

of c (except possibly at c), i.e., there exists $\delta > 0$, s.t. $\underbrace{(c-\delta, c+\delta) \setminus \{c\}}_{\substack{\text{i.e.,} \\ f \text{ may not be defined at } c}} \subseteq D$
 $= (c-\delta, c) \cup (c, c+\delta)$

Def. Let A and B be sets. The set A minus B is the set $A \setminus B \triangleq \{x \in A : x \notin B\}$.

E.g.



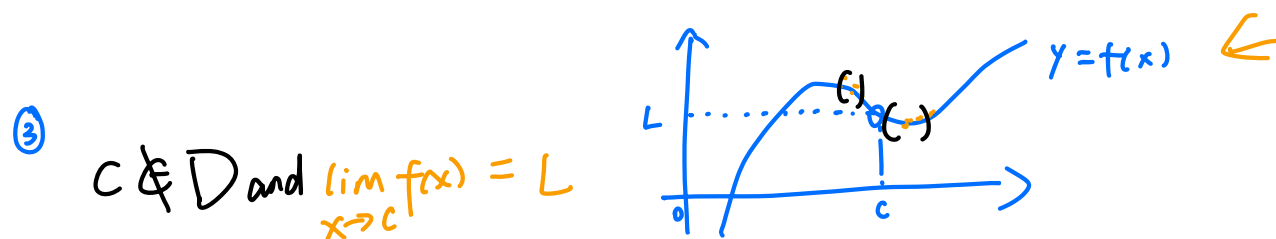
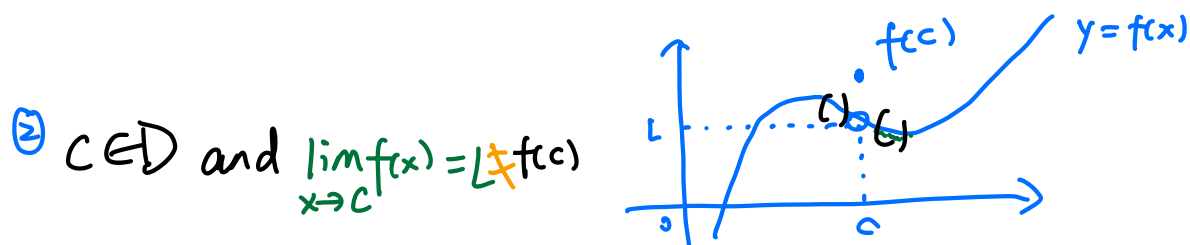
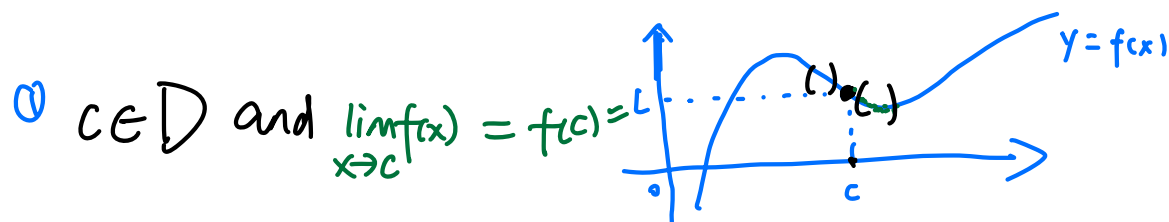
$$(c-\delta, c+\delta) \setminus \{c\} = (c-\delta, c) \cup (c, c+\delta).$$

§4.3. finite limit

Let $c \in \mathbb{R}$ (so $L \neq \pm\infty$). The symbol $\lim_{x \rightarrow c} f(x) = L$ is read as

"the limit of $f(x)$ as x approaches c is L ."

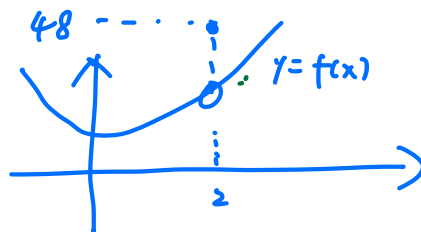
e.g. For all graphs below, we have $\lim_{x \rightarrow c} f(x) = L$.



Rmk. $\lim_{x \rightarrow c} f(x)$ measures the "trend" of f as x approaches c .

But in general, $\lim_{x \rightarrow c} f(x)$ is not always equal $f(c)$.

Eg for ②. $f(x) = \begin{cases} x^2 + 1, & \text{if } x \neq 2. \\ 48, & \text{if } x = 2. \end{cases}$
 $f(2) = 48$



Compute: $\lim_{x \rightarrow 2} f(x) = \lim_{\substack{\Delta x \rightarrow 0 \\ x = 2 + \Delta x}} [(2 + \Delta x)^2 + 1] = \lim_{\Delta x \rightarrow 0} [4 + 4\Delta x + (\Delta x)^2 + 1]$
 $= \lim_{\Delta x \rightarrow 0} [5 + 4\Delta x + (\Delta x)^2] = 5.$

Eg for ③ (in §5.1). $f(x) = \frac{x^2 - 1}{x - 1}$, $x \in (-\infty, 1) \cup (1, \infty)$ [$1 \notin D$].

Compute: $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{\substack{\Delta x \rightarrow 0 \\ x = \Delta x + 1}} \frac{(\Delta x + 1)^2 - 1}{\Delta x + 1 - 1} = \lim_{\Delta x \rightarrow 0} \frac{(\Delta x)^2 + 2\Delta x + 0}{\Delta x}$
 $= \lim_{\Delta x \rightarrow 0} (\Delta x + 2) = 2.$

Comments. Examples are important. Math theory are originated from and generalized from concrete examples; instead of coming out from nowhere.

§4.4. Formal Definition. [Optional]

Def. Let $f: D \rightarrow \mathbb{R}$ be a function defined on an open interval containing c , except possibly at c itself (i.e., c may be not in the domain like case ③).

Let $L \in \mathbb{R}$. Then we say "the limit of f at c exists and equals L "

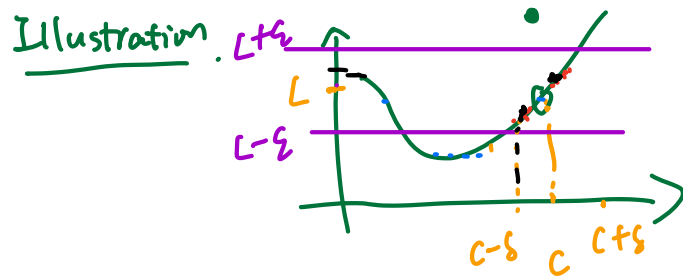
and write $\lim_{x \rightarrow c} f(x) = L$ if for any $\varepsilon > 0$, there exists a $\delta > 0$ such that for any $x \in D$

with $0 < |x - c| < \delta$, we have $|f(x) - L| < \varepsilon$.

(i.e. $x \in (c - \delta, c) \cup (c, c + \delta)$)

$f(x) \in (L - \varepsilon, L + \varepsilon)$
: $f(x)$ is close to L in a neighborhood.

"in the neighborhood, all values coincide the trend".



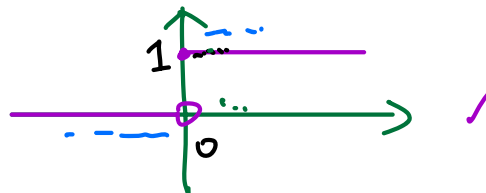
① Let ε be any arbitrary small value (e.g., $\varepsilon = 0.01$)

② Find a $\delta > 0$, s.t. $|f(x) - L| < \varepsilon$ on $x \in (c - \delta, c + \delta) \setminus \{c\} \cap D$.

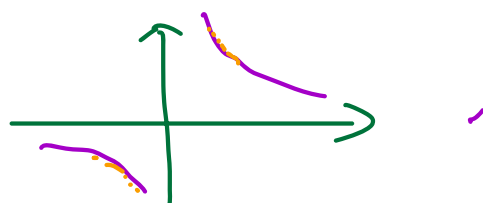
• Go back and examine the above three graphs.

§4.5. counter examples: $\lim_{x \rightarrow 0} f(x)$ does not exist.

Case 1. $f(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$



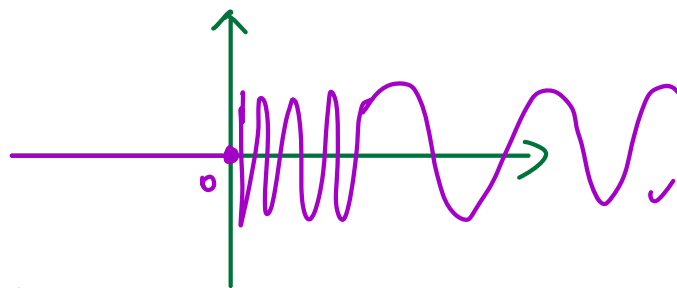
Case 2. $f(x) = \begin{cases} \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$



Case 3. $f(x) = \begin{cases} 0, & x \leq 0 \\ \sin \frac{1}{x}, & x > 0 \end{cases}$

$\frac{1}{0.001}, \frac{1}{0.002}$

$\sin [1000, 500]$



Reasons:

Case 1: approaching different numbers from different sides:

$f(x) \rightarrow 0$ as $x \rightarrow 0$ from left.

$f(x) \rightarrow 1$ as $x \rightarrow 0$ from right.

[Formally, suppose $\lim_{x \rightarrow 0} f(x)$ exists and equals L . Then L should be 0 or 1.

Take $L = 0$ as an example. We set $\epsilon = 0.5$.

Assume by contradiction that ^(there exists) $\exists \delta > 0$ ^{st.} $|f(x) - 0| < \epsilon$ for any $x \in (0 - \delta, 0) \cup (0, 0 + \delta)$.

But it is **not true** because if $x \in (0, 0 + \delta)$, then $|f(x) - 0| = |1 - 0| = 1 > 0.5$.

It is a contradiction!

Hence, $\lim_{x \rightarrow 0} f(x)$ does not exist.]

Case 2: As x approaches 0, $f(x)$ can be arbitrarily large or small
[so L is not finite].

Case 3. Oscillating from the right between two numbers, $(-1, 1)$

i.e. At any neighborhood of 0, there exists two points x_0 and x_1 ,

$$\text{s.t. } \begin{cases} |x_0 - 0| < \delta \text{ and } |x_1 - 0| < \delta \\ \sin(\frac{1}{x_0}) = 1 \text{ and } \sin(\frac{1}{x_1}) = -1 \end{cases} \quad \left[\begin{array}{l} \frac{1}{x_0} = \frac{\pi}{2} + 2k\pi, \Rightarrow x_0 = \frac{1}{\frac{\pi}{2} + 2k\pi}, k=1,2,3,\dots \\ \frac{1}{x_1} = -\frac{\pi}{2} + 2k\pi \Rightarrow x_1 = \frac{1}{-\frac{\pi}{2} + 2k\pi}, k=1,2,3,\dots \end{array} \right]$$

making $|\sin(\frac{1}{x_0}) - L| = |1 - L| < \varepsilon$ and $|\sin(\frac{1}{x_1}) - L| = |-1 - L| < \varepsilon$

not hold at the same time.

Alternative notation. We can write $f(x) \rightarrow L$ as $x \rightarrow c$ to represent $\lim_{x \rightarrow c} f(x) = L$.